Period 5 Self Eval Writeup for NASA Openscapes (8/1/25-7/31/26)

During Period 5, the NASA Openscapes team continued to serve as a main access point for NASA Earth science and tutorial development on the Cloud. By working with the DAAC staff who participate as mentors, the team continued to provide open, collaborative, reproducible work aligned with NASA goals to enable Open Science. The NASA Openscapes team supports data usage enablement through Open Science with a focus on using commercial cloud through training, mentoring, and community engagement - by building a teaching community across the NASA Distributed Active Archive Centers (DAACs) and collaborating with the broader open science community.

During Period 5, we focused on strengthening NASA networks between DAACs, missions, training programs, and applications, aligned with SPD-41a Open Science, as we continued to support users through developing skills and tools. Specific focus areas for this period, co-developed with the Mentors:

- User support and co-development
- *earthaccess* python library
- Champions Cohort (Fall 2025)
- Earthdata Cloud Cookbook

Major activities completed by the Openscapes team during Period 5 are highlighted below.

Completing All Deliverables.

The Openscapes team worked efficiently and completed deliverables called out in the Statement of Work (SOW). They include 6 "Imagine What's Possible" webinars, 4 Coding Skills workshops, and 2 NASA-Specific Data workshops to different scientific and data audiences. They are listed below.

"Imagine What's Possible" Webinars:

1. Mentor Danny Kaufman (ASDC) developed and led an ASDC Access Mini-Workshop, which included inspiring talks and experimentation to demonstrate virtual dataset capabilities for satellite data analysis. There were 9 participants, who helped give feedback and ideas for next steps forward in August, 2025.
2. Mentors Amy Steiker (NSIDC), Danny Kaufman (ASDC), and Julie Lowndes (Openscapes) co-presented an invited talk entitled "Crossing the chasm at NASA Earthdata and NOAA Fisheries" in the virtual seminar series at NCAR CISL on September 25, 2025. There were 21 attendees.

3. Julie Lowndes (Openscapes) received the AGU Dr. Greg Leptoukh Award Lecture Award “Recognizing Computational and Data Advances that Enable Scientific Discovery”, for which she was nominated by the NASA Mentors. Her 75-minute talk at AGU in December 2025 was titled “Forking as a worldview: How Openscapes uses the open source concept of copy-modify to support researchers & shift culture in science”, and she also gave a “Reprise” lecture online in March 2026, organized by the AGU Informatics section.
4. Mentor Amy Steiker (NSIDC) and Julie Lowndes (Openscapes) presented The extensible social infrastructure enabling NASA Earthdata science in the cloud. This was one of the 10 talks and posters the NASA Openscapes Mentors led or co-authored at the AGU Fall Meeting in New Orleans in December 2025. This was an invited talk in a session called Open-Source Geospatial Workflows in the Cloud: Tools and Techniques for Data Access, Analysis, Visualization, Storytelling, and Sharing in the Python and Jupyter Ecosystem II Oral, and there were > 100 in person attendees. Planning for all AGU talks is documented at github.com/NASA-Openscapes/2025-agu-planning.
5. Mentor Amy Steiker (NSIDC) and Julie Lowndes (Openscapes) gave an invited talk entitled "Openscapes: The extensible technical and social infrastructure enabling NASA Earthdata science in the cloud" at the USGS Data Practitioners Meeting on March 18, 2026, focused on virtualizing data and NASA Openscapes progress. There were 30 attendees.
6. Mentors Luis Lopez (NSIDC), Amy Steiker (NSIDC), Danny Kaufman (ASDC), Justin Rice (ESDIS), and Julie Lowndes (Openscapes) will present talks in a 90-minute session they organized titled “earthaccess: five years of open source science leadership at NASA” at the ESIP Summer Meeting in Austin, Texas on July 29, 2026. Two other sessions Openscapes organized and Mentors co-lead: "Facilitating scientists' learning while 'stress testing' open science principles via Openscapes"; and "Bridging Divides and Getting to Yes."

Coding Skills Workshops:

1. Openscapes Mentors Michele Thornton and Rupesh Shrestha (ORNL DAAC) led a workshop entitled "Plot to Plane: Working with NASA and NEON Airborne and Field Datasets" at the annual Ecological Society of America (ESA) conference on August 15, 2025. There were 20 participants.
2. Andy Barrett (NSIDC) developed and taught "ICESat-2 Sea Ice Freeboard during the MOSAiC Expedition" as part of the Fall 2025 Champions Cohort on November 13, 2025, with a hands-on review in the 2i2c JupyterHub on November 14. There were 30 participants.
3. The team organized and supported the earthaccess December 15 Hackday in New Orleans and online, supported through ESIP's FUNding Friday initiative and co-hosted by the Cloud Native Geospatial Forum (CNG) and Openscapes. The goal was to build a prototype and roadmap for "Growing the Family" by integrating earthaccess with

non-NASA Earth science data (NOAA, USGS, Copernicus). We organized across four contributor tracks with people in person and online. There were 40 participants.

4. The team is organizing and hosting the ESIP 2026 Summer Meeting Monday Hackday, themed "Making Things Easier for Scientists", on July 27, 2026. This is focused on supporting scientists with infrastructure and surfacing the technical and social friction in cross-agency data access. The schedule page is here; we organized this as a pre-ESIP event following our FUNding Friday hackday.

NASA Specific Workshops:

1. ORNL DAAC Mentors Michele Thornton and Rupesh Shrestha led "Airborne Data Applications for Invasive Species Mapping", hosted by NASA Applied Remote Sensing Training (ARSET), on September 30, 2025. Hands-on exercises were executed on the Openscapes 2i2c JupyterHub. There were 401 participants from 68 countries.
2. Mikala Beig (NSIDC) presented the NASA Earthdata Webinar Short "Downloading Data from the NASA Earthdata Cloud Using the earthaccess Python Library" on April 22, 2026. There were 50+ attendees. The recorded short can now introduce earthaccess during hackdays and workshops.

Managing & Supporting Activities in Coordination with the DAACs:

- Hosted biweekly Mentor Calls with agendas co-designed with DAAC Mentors throughout the period. Topics included AGU abstract submissions, Champions Cohort scoping, Cookbook visioning, earthaccess governance, virtual datasets and development with industry, cross-government, and academic partners to support NASA Earthdata users, restructuring, and potential integration with Web Unification, and developing coaching skills with NOAA Fisheries mentors.
- Ran the **Fall 2025 Champions Cohort** (link) with 30 participants out of 130 registrations. Lower-than-expected participation was attributed to the government shutdown. We designed and led this as a shorter event that was more viable for Mentors, who wanted to support users and recognized both users and mentors had limited availability.
- Hosted recurring **Hackdays**: *earthaccess* (biweekly); Cookbook (biweekly)
    - **earthaccess hackdays;** in addition to open governance decisions and development, the Mentors advanced preparation of the library for publication in the Journal of Open Source Software (PR), now undergoing pre-review via pyOpenSci (submission).
    - **Cookbook coworkings** focused on visioning, restructuring and deepening community-based governance, including a Cookbooks Style Guide and Author Style Guide.
- Added **53 of ESDIS's top datasets to the AWS Open Data Registry** via a merged pull request, with help from ESDIS Summer Interns Ethan Grant and Kasey Wei.
- Documented **2026 Mentor Goals** at nasa-openscapes.github.io/mentors.html#goals.

- Collaborated at the **Connected Data Products Hackdays** (April 13-15, 2026, UC Berkeley; event page), with Joe Kennedy (ASF), Luis López (NSIDC), and Julie Lowndes joining Tasha Snow, Fernando Perez, and others.

Maintaining Web Presence and Providing Web-Accessible Resources

The Openscapes team is maintaining the web presence and the web accessible resources through the current Openscapes website (openscapes.org), as well as the NASA project website https://nasa-openscapes.github.io, with active contributions to the NASA Earthdata Cloud Cookbook https://nasa-openscapes.github.io/earthdata-cloud-cookbook. During this period the team undertook a substantial Cookbook visioning and restructuring effort - streamlining the table of contents and navigation, reorganizing and cleaning up content, and identifying maintenance issues - so that Mentors can focus on content updates and maintenance. The Openscapes team is implementing much of the restructuring the Mentors identified, with work continuing through the summer. The team also advanced Cookbook R tutorials, including CMR-STAC and TEMPO with R. Additionally, the *earthaccess* python library was added to the NASA Earthdata website: https://www.earthdata.nasa.gov/data/tools/earthaccess, as well as the Cookbook: https://www.earthdata.nasa.gov/learn/tutorials/earthdata-cloud-cookbook.

Writing and Publishing Lessons Learned from NASA Openscapes.

The Openscapes team collaborated across organizations and published lessons learned based on NASA Openscapes work. Highlights include:

- Published the peer-reviewed article "A path to better science through co-creation and open infrastructure" in *Earth and Space Science*, co-authored across NASA Openscapes Mentors (Luis López, NSIDC; Joe Kennedy, ASF), the Openscapes team (Julie Lowndes, Andy Teucher), 2i2c (Yuvi Panda, James Colliander), NOAA Fisheries (Eli Holmes), the NSIDC UWG (Jessica Scheick), UC Berkeley (Carl Boettiger), and Development Seed (Alex Mandel).
- Published a blog post on the earthaccess December 15 Hackday, co-authored by 22 co-leads and facilitators from six NASA DAACs, Development Seed, UC Berkeley Center for Data Science & Environment, the University of New Hampshire, ESIP, and Openscapes.
- Contributed to 2i2c blog posts documenting cloud cost work developed and tested in the Openscapes JupyterHub: transparent cloud cost monitoring with user-level dashboards, JupyterHub SSH, and a shared-password admin workflow to ease workshop leads acting as admins.
- Advanced a cross-NASA JupyterHub whitepaper on lessons learned, led by Tasha Snow, with the draft nearing completion.
- Prepared the *earthaccess* software paper for the Journal of Open Source Software (now in pyOpenSci pre-review).

- Monthly cost reports as downloadable PDFs from our NASA Openscapes JupyterHub (as well as NOAA Fisheries Openscapes) are posted here: https://openscapes.cloud/usage-reporting.

NASA Openscapes KPIs Definition and KPIs Achieved

The Openscapes team has defined key performance indicators (KPIs) with which to evaluate the progress of the Enabling Science in the Cloud initiative including the impact on targeted communities. They are posted and maintained at github.com/NASA-Openscapes/how_we_work.

The KPIs are grouped under two categories - "Teaching Numbers" and "Engineering Numbers". The KPIs achieved for Period 5 are included in the table below.

KPI Definition	**Notes**	**KPI Achieved (8/1/25-7/31/26)**
Teaching Numbers		
Number of Cloud participants in the Workshop JupyterHub	This reflects the number of participants in hands-on workshops hosted by the NASA Mentors	419 unique usernames in the Workshop hub since Aug 1 2025.
Number of new teaching slides and tutorials	This reflects the Champions program lessons as well as new tutorial books. Note that we reuse lessons across workshops.	8
Number of talks "imagine what's possible" conducted	This reflects talks that are not hands-on (and do not provide access to JupyterHub)	7
Engineering Numbers		
Number of contributors to earthaccess	This is currently taken from GitHub contributor count	59
Number of dependents on earthaccess	This is taken from the GitHub "used by" count	270
Number of stars for earthaccess	This is taken from the GitHub count of users that opt-in to	613

KPI Definition	Notes	KPI Achieved (8/1/25-7/31/26)
	"starring" earthaccess and saving it to their favorites list	
Number of contributors to Cookbook	This is also taken from GitHub contributor count	42
Number of contributors to GitHub docker image	This is also taken from GitHub contributor count	14
Number of items in workshop-planning GitHub Issues (open cross-DAAC planning)	This shows activities proposed and pursued across the community. It does not distinguish between which were completed versus considered but gives an idea of scope	174
Number of cloud infrastructure hack days (2i2c, environments, etc)	This gives an idea of engagement and regular progress across the NASA Openscapes community goals	6

Value: NASA Openscapes is the main access point for NASA Earth science & tutorial development on the Cloud. The EED-3 Openscapes team provides a critical service of building the cross-DAAC mentor community with a shared user-centered teaching & development approach that is networked with the broader open science community and further coordinates, role-models, and teaches Open Science through technical and leadership skill building via Mentors and Champions programs, to help ESDIS achieve NASA's goals of data usage enablement through Open Science.